Robust Lane Detection Under Varied Lighting Conditions

Nishant Suresh
Kateryna Lysytsyna
Sagar Kaistha

The Problem

- Lane detection is critical for autonomous driving safety
- Lane departures cause ~50% of fatal car crashes (13,000–19,000 deaths/year in US)
- Performance degrades under challenging conditions: nighttime, shadows, glare, occlusions
- Challenge: Achieve robust detection across all real-world scenarios

CULane Dataset

- 133,235 frames (88,880 train, 9,675 val, 34,680 test) collected in Beijing by CUHK & SenseTime. Resolution: 1640×590.
- 9 scenarios: Normal, Shadow, Curve, Highlight, Crowded, Arrow, No Line, Night, Intersection
- Per-pixel segmentation labels (0=background, 1–4=lanes) enable pixel-wise evaluation
- Preprocessing:
  - Resize to 512×256
  - Normalize to [0,1]
  - Convert masks to binary (0=background, 1=lane)

Challenges

- Low contrast at night.
- Shadow boundaries create uneven illumination.
- Glare washes out lane markings.
- Severe class imbalance (~0.5% lane pixels).

Proposed Models

Task: Pixel-wise binary classification (lane vs. background)

Model	Type	Input	Library
Logistic Regression	Baseline	326 HOG features from 32×32 patch	scikit-learn
Lightweight CNN	Shallow DNN	3×256×512 RGB image	PyTorch
U-Net	Deep DNN	3×256×512 RGB image	PyTorch

Evaluation & Expected Outcomes

- Metrics: F1 score based on IoU (TP: IoU ≥ 0.5, FP: IoU < 0.5)
- IoU: Intersection of actual and predicted lanes / Union of the 2 lanes
- Per-scenario evaluation to assess robustness across lighting conditions